Applied NLP

DATA 641

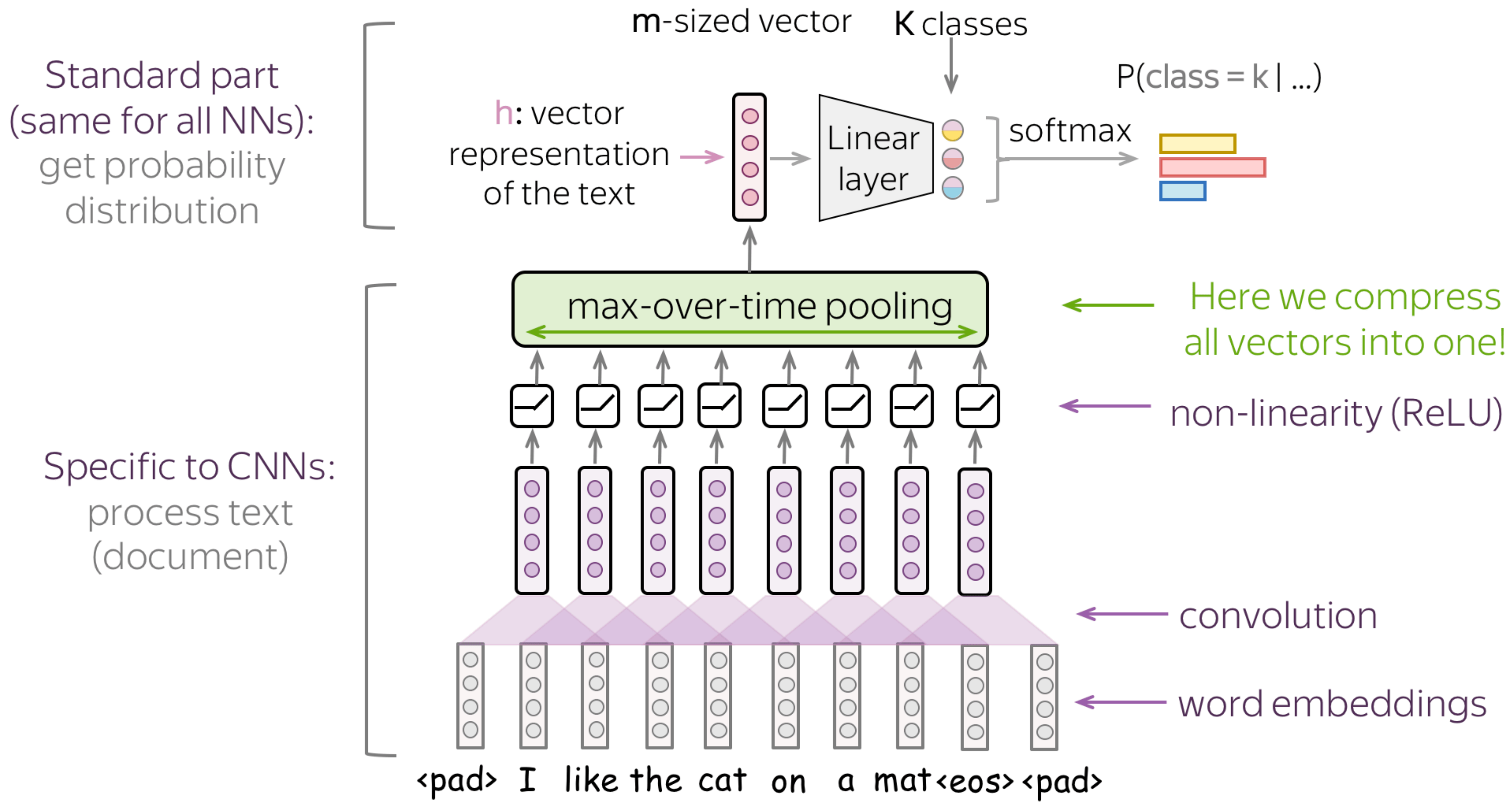
Lecture 7 — CNNs for text classification

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action, Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, First edition, MANNING

*Figures are from the lecture notes: https://lena-voita.github.io/nlp_course

Basic convolutional model for text classification

After the convolution, we use global-over-time pooling. This allows to compress a text into a single vector. The model itself can be different, but at some point, it has to use the global pooling to compress input in a single vector.



Several convolutions with different kernel sizes